ABAID ULLAH BHATTI

Software Engineer • Full-Stack Developer • Backend & Systems Specialist

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EDUCATION

University of Engineering and Technology (UET), Lahore — Lahore, Pakistan

Bachelor of Science in Computer Science (7th Semester) — CGPA: ***3.351 / 4.00*** — *2023 – 2027*

- **Academic Scholarship:** Awarded Government Merit-Based Academic Scholarship (2020 – 2025).
- **Core Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Operating Systems, Computer Networks, Software Engineering, Theory of Automata, Compiler Construction.

TECHNICAL SKILLS

Languages:	TypeScript, JavaScript (ES6+), Python, C, C++, SQL, x86 Assembly (MASM32), HTML5, CSS3
Frontend:	React 18, Next.js 16 (App Router), Tailwind CSS, Vite, Redux Toolkit, Component Architecture
Backend & APIs:	Node.js, Express.js, ASP.NET Core, RESTful APIs, JWT, Role-Based Access Control (RBAC), Socket.IO, Kafka
Databases & Cloud:	PostgreSQL, Prisma ORM, MongoDB (Atlas/Mongoose), SQL Server, Redis, Supabase, Cloudinary, Docker
Tools & Testing:	Git, GitHub, Docker, Postman, Newman, Jest (58+ suites), Vitest, Supertest, Linux / Bash, CI/CD, Markdown
Engineering Focus:	Event-Driven Architecture, High-Concurrency DB Locks, AST Compilers, Stripe Payment Pipelines

WORK EXPERIENCE

Software Engineering Intern — Quaid Ventures — June 2026 – August 2026

Backend & Full-Stack Engineering • Verified Experience Letter — *[Verified Letter on GitHub ↗]*

- Conducted targeted debugging and QA on **Shipr**, the company's flagship platform, resolving complex AI-related bugs to improve system reliability and data accuracy.
- Built an event-driven, asynchronous backend task system using **Apache Kafka** for background processing and **Redis** for real-time analytics, caching, and granular role-based access control (RBAC).
- Engineered a high-concurrency **Inventory Management System** using **PostgreSQL** and **Prisma**, applying row-level locks and atomic transactions to eliminate race conditions and overselling.
- Developed **Shop.co**, a production-ready full-stack clothing commerce application with React, Redux, Axios, Stripe checkout transactions, and MongoDB analytics.
- Architected a modular backend for an **Enterprise HR Portal** spanning 19 business domains (workforce management, leave workflows, attendance, and payroll).
- Built a full-duplex real-time chat and messaging engine using **Socket.IO** and **Kafka**, enabling JWT-authenticated private and group messaging.

FLAGSHIP ENGINEERING PROJECTS

Litho: Markdown-to-LaTeX Publishing Ecosystem (Final Year Project) — Python, AST Compiler, Tectonic TeX, Jinja2, CI/CD

Compiler & LaTeX Backend Engine Lead — Department of Computer Science, UET Lahore — *2024 – Present*

- Architected an end-to-end technical publishing compiler mapping CommonMark AST nodes into versioned publisher LaTeX templates with lossless round-trip metadata recovery.
- Engineered a custom Jinja2 template-binding emitter generating high-fidelity LaTeX for callouts, math equations, citations, figures, and Gantt charts.
- Configured and packaged a trimmed Tectonic TeX engine fork with precompiled format files, accelerating PDF compilation times by over 40%.
- Built an automated golden-PDF visual regression testing harness in CI, verifying pixel-perfect diff rendering across IEEE, Springer, and thesis templates.

KEY ENGINEERING PROJECTS (CONTINUED)

Shop.co — Enterprise MERN Stack E-Commerce Platform — **React 18, TypeScript, Node.js, Express, MongoDB, Stripe**
Full-Stack Architect & Developer — Live on Vercel & Render Cloud — *2024*

- Engineered a production-ready, fully-typed clothing commerce application with strict RBAC authorization and JWT authentication with access/refresh token rotation.
- Implemented unified shopping cart persistence supporting guest UUID sessions and automatic cart merging upon user authentication.
- Leveraged atomic MongoDB transactions to guarantee inventory consistency during concurrent multi-item checkouts and Stripe webhook fulfillment.
- Built an admin analytics dashboard with native MongoDB aggregation pipelines tracking revenue, product velocity, and customer cohort metrics.

Litho Template Marketplace — Curated Document Registry — **Next.js 16, React, Express 5, Supabase PostgreSQL, Docker**
Full-Stack Developer — Live on Render Cloud Docker Container — *2024*

- Developed a responsive template registry featuring dual-pane split view, real-time multi-token search (tag:, kind:), and a 10-theme CSS design system.
- Integrated Supabase PostgreSQL database with schema migration pipelines and built administrative moderation controls for submissions.

Cross-Database Performance Benchmarking Suite — **Python, PostgreSQL, MongoDB, SQL Server, Redis, Matplotlib**
Solo Developer & Benchmarking Researcher — *2024*

- Engineered an automated Python benchmarking framework measuring throughput and latency for CRUD, multi-table JOINs, and complex aggregation operations.
- Benchmarked SQL Server, PostgreSQL, MongoDB Atlas, and Redis under concurrent workloads, analyzing index optimization and query execution plans.

Attendance Management System (AMS) — **Next.js 14, React, Supabase PostgreSQL, Tailwind CSS**
Full-Stack Web Engineer — *2024*

- Built an administrative attendance tracking platform with role-based access for faculty and administrators, daily attendance logging, and CSV exports.

OPEN-SOURCE & LEADERSHIP EXPERIENCE

Solo Full-Stack Engineer / Open-Source Contributor — **Remote / Lahore, Pakistan**
Independent Software Development (11 Repositories on GitHub) — *2024 – Present*

- Architected and deployed 11 open-source repositories spanning full-stack web platforms, compiler toolchains, cybersecurity scanners, and low-level systems.
- Engineered production-grade security architectures including JWT authentication, rate limiting, helmet headers, and comprehensive unit/integration test suites with Jest.

UET ACM Student Chapter — **UET Lahore, Pakistan**
Event Coordinator & Technical Volunteer — *2023 – Present*

- Coordinated collegiate technical workshops, coding bootcamps, and hackathons for 300+ computer science students.

VERIFIED CERTIFICATIONS & TECHNICAL CREDENTIALS

- **Generative AI Application Developer (Pak Angels, Silicon Valley, CA):** Awarded *Top Performer* in GenAI training (June–July 2025). Certificate ID: `322056cbc677032d`. [Verified Credential on GitHub ↗]
- **IT Department Internship Certificate (Tracking):** Completed 2-month professional software engineering internship (August 2026). [Verified Certificate on GitHub ↗]
- **Full-Stack Web Development:** Comprehensive single-page application architectures, Next.js / React, and RESTful Express APIs (2024).
- **Relational Databases & SQL:** Advanced query optimization, schema normalization (1NF–BCNF), and ACID transactions (2024).
- **Python Programming & Scripting:** Object-oriented design, systems automation, and data pipeline scripting (2024).
- **MongoDB & Document Databases:** Aggregation pipelines, Atlas cluster administration, and Mongoose schema modeling (2024).
- **Modern JavaScript (ES6+) & Data Structures (DSA):** Asynchronous event loop, closures, and algorithmic problem solving (2023).